

Evaluating Deep Speaker Embedding Robustness to Domain, Sampling Rate, and Codec Variations

Alexandre Ferro Filho¹, Diogo Fernandes Costa Silva¹, Pedro Elias Engelberg Silva Borges¹,
Arlindo Rodrigues Galvão Filho¹

¹AKCIT, Federal University of Goiás, Brazil

diiogofernands@gmail.com

Abstract

Speaker verification systems based on deep speaker embeddings perform well under matched training and evaluation conditions, but performance degrades under domain shifts. This work evaluates the sensitivity of four speaker embedding models—ECAPA-TDNN, TitaNet, ECAPA2, and ReDimNet—to variations in acoustic domains, sampling rates, and audio codecs. Experiments on datasets with far-field speech, noise, and music interference demonstrate that all models degrade under mismatched conditions. ReDimNet demonstrates the smallest performance degradation compared to the other models but is still affected in certain cases. Downsampling and low-bitrate compression further degrade performance, revealing the reliance of models on high-frequency information and their sensitivity to compression artifacts.

Index Terms: speaker verification, speech recognition, computational paralinguistics

1. Introduction

Automatic Speaker Verification (ASV) systems are deployed in biometric authentication, access control, and voice assistants, requiring reliable performance across diverse environments. While deep speaker embedding models, such as x-vector [1], ECAPA-TDNN [2], TitaNet [3], and ReDimNet [4], achieve high accuracy under matched training and evaluation conditions, their performance degrades under domain shifts.

Variations in channel characteristics, codecs, sampling rates, background noise, and microphones introduce shifts that significantly reduce ASV accuracy. Models trained on clean, wideband datasets like VoxCeleb [5] often fail in narrowband, noisy, or far-field conditions [6]. Approaches such as domain adversarial training [7], and data augmentation [8, 1] improve robustness but remain limited when facing unseen domains or extreme distortions [9, 10].

Recent datasets expose these weaknesses. CN-Celeb [11] captures multi-genre, real-world speech with noise; DiPCo [9] simulates far-field meeting scenarios with reverberation and overlapping speakers; JuKeBOX [12] introduces low-quality speech mixed with music. Although these datasets highlight performance gaps, the extent to which state-of-the-art models degrade across domain shifts, sampling rates, and codec variations remains underexplored.

This work investigates the sensitivity of deep speaker embeddings to domain shifts. We evaluate ECAPA-TDNN, TitaNet, ECAPA2, and ReDimNet across far-field (DiPCo), multi-genre (CN-Celeb), noisy and reverberant (VOICES), and low-quality music-mixed speech (JuKeBOX). Our analysis reveals performance degradation across domains and highlights the need for more robust embedding learning strategies.

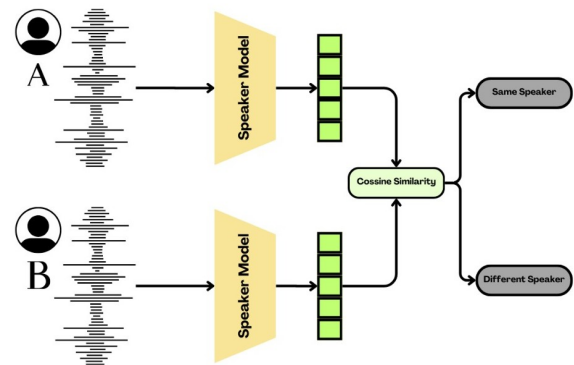


Figure 1: Illustration of the speaker verification pipeline. Audio signals from two speakers are processed by a speaker embedding model to extract deep speaker representations. Cosine similarity is computed between the embeddings to determine whether the samples correspond to the same or different speakers.

Our key contributions are summarized as follows:

- We systematically evaluate the performance degradation of deep speaker embeddings under domain shifts, quantifying the impact of diverse acoustic conditions on state-of-the-art models.
- We assess model sensitivity across diverse evaluation datasets, covering far-field (DiPCo), multi-genre broadcast (CN-Celeb), noisy and reverberant (VOICES), and low-quality music-embedded speech (JuKeBOX), showing that VOICES exposes notable weaknesses in noise robustness.
- We benchmark four leading speaker verification models—ECAPA-TDNN, ECAPA2, TitaNet, and ReDimNet—and find that ReDimNet consistently exhibits the lowest degradation across domains, but still suffers in music-contaminated and low-quality conditions.
- We analyze the effects of sampling rate reduction, and low-bitrate codecs, revealing that all models are sensitive to downsampling and compression artifacts, with ReDimNet being the least affected.

2. Related Works

Speaker verification systems have advanced through improvements in deep speaker embedding architectures, domain adaptation techniques, and evaluation across diverse acoustic conditions. The x-vector model [1] introduced TDNN-based embeddings, while ECAPA-TDNN [2] incorporated channel attention,

context gating, and multi-scale aggregation. ResNet-based architectures, such as ECAPA2 [13] and TitaNet [3], further enhanced embedding extraction with squeeze-excitation and attentive pooling. ReDimNet [4] proposed a more efficient design optimized for both clean and noisy data. These models achieve competitive performance in matched conditions, but their robustness degrades under domain shifts, indicating that architectural refinements alone are insufficient.

To address this, recent works have explored leveraging pretrained speech models and domain adaptation strategies. Whisper-SV [14] adapts the Whisper ASR model for speaker verification using multi-layer aggregation and attention, improving performance in low-resource settings. Transfer learning and knowledge distillation from ASR-trained Conformers [15] reduce EER in VoxCeleb evaluations. Multi-Level Domain Adapters [6] introduce domain-specific modules that encode domain information into embeddings, achieving up to 21.1% EER reduction on the 3D-Speaker dataset. While these methods improve robustness, they often rely on domain labels, large-scale pretraining, or target-domain adaptation, limiting their applicability across unseen domains.

Despite these advances, domain mismatch remains a fundamental challenge. Models trained on clean, wideband datasets like VoxCeleb [5] degrade in narrowband, noisy, or far-field conditions. Early work explored domain adversarial training [7] and data augmentation [8], but their effectiveness diminishes under severe acoustic shifts [7, 8]. Evaluation datasets such as CN-Celeb [11], DiPCo [9], VOICES [16], and JuKeBOX [12] expose these limitations, highlighting the performance gap across diverse domains. However, prior work has not systematically isolated the combined impact of sampling rates, codecs, and channel characteristics on embedding robustness. Our work fills this gap by quantifying the performance degradation under these factors, offering a more comprehensive assessment of embedding resilience.

3. Experimental Setup

3.1. Datasets

We evaluate model performance using four datasets selected to expose the sensitivity of speaker embeddings to diverse domain shifts. CN-Celeb [11] contains multi-genre speech with varied recording conditions and noise levels, reflecting channel and content variability typical of media broadcasts. VOICES [16] comprises far-field speech with noise and reverberation, assessing robustness to distance and acoustic interference. DiPCo [9] features multi-party conversations with speaker overlap and room reverberation, representing meeting scenarios with diarization challenges. JuKeBOX [12] consists of speech mixed with music and compression artifacts, testing resilience to non-speech interference and low-quality signals. These datasets collectively cover critical domain variations, enabling a detailed analysis of embedding degradation across conditions encountered in real-world applications.

3.2. Metrics

We evaluate speaker verification performance using Equal Error Rate (EER) and minimum Detection Cost Function (minDCF), two principal metrics in the field. EER represents the threshold at which the False Acceptance Rate (FAR) equals the False Rejection Rate (FRR), providing a threshold-independent assessment. minDCF considers application-specific costs, weighting missed detections and false acceptances based on prior proba-

bilities.

EER is defined as the operating point where:

$$\text{FAR}(\tau) = \frac{\text{False Acceptances}}{\text{Non-target Trials}}, \quad \text{FRR}(\tau) = \frac{\text{False Rejections}}{\text{Target Trials}} \quad (1)$$

and

$$\text{EER} = \text{FAR}(\tau_{\text{EER}}) = \text{FRR}(\tau_{\text{EER}}) \quad (2)$$

where τ_{EER} is the threshold where the rates are equal.

The detection cost function (DCF) is defined as:

$$\text{DCF}(\tau) = C_{\text{miss}}P_{\text{miss}}(\tau)P_{\text{target}} + C_{\text{fa}}P_{\text{fa}}(\tau)(1 - P_{\text{target}}) \quad (3)$$

where $P_{\text{miss}}(\tau)$ and $P_{\text{fa}}(\tau)$ are the miss and false acceptance probabilities, and C_{miss} , C_{fa} , and P_{target} represent the costs and prior. The minimum DCF is:

$$\text{minDCF} = \min_{\tau} \text{DCF}(\tau) \quad (4)$$

3.3. Models

We evaluate four speaker embedding models commonly adopted in speaker verification: ECAPA-TDNN, TitaNet, ECAPA2, and ReDimNet. These models incorporate different architectural choices aimed at improving robustness and discriminability in embedding extraction. Table 1 summarizes the domains and augmentation techniques applied during training.

ECAPA-TDNN [2] builds upon the x-vector architecture, introducing multi-scale feature aggregation, squeeze-excitation blocks, and channel-wise attention to enhance speaker representation. ECAPA2 [13] refines this design with additional modifications to the network structure, maintaining the same training data and augmentation strategies, which include noise, music, babble, reverb, speed perturbation, and compression applied to wideband, interview-style speech.

TitaNet [3] also derives from the TDNN framework but introduces residual connections and attention mechanisms, focusing on scalability and efficiency. Unlike the ECAPA models, TitaNet was trained on a broader range of domains, including telephone and read speech, with data augmentation involving reverb, speed perturbation, and time/frequency masking.

ReDimNet [4] diverges by prioritizing robustness to domain shifts. It employs regularization techniques and discriminative training objectives to mitigate the impact of channel variability and noise. Its training data aligns more closely with ECAPA models, consisting of wideband, interview-style speech augmented with noise, music, babble, reverb, and speed perturbation.

4. Methodology

We investigate the domain sensitivity of deep speaker embeddings by evaluating the performance of ECAPA-TDNN, ECAPA2, TitaNet, and ReDimNet under variations in acoustic domains, codecs, and sampling rates. The evaluation follows the speaker verification pipeline shown in Figure 1, where audio samples are processed by a model to extract speaker embeddings, and cosine similarity is computed between pairs to determine whether they belong to the same speaker. A balanced set of test trials was created for each evaluation dataset, ensuring an equal number of same-speaker and different-speaker pairs.

We select evaluation datasets that represent domains distinct from those used in model training to expose performance degradation under domain shifts. CN-Celeb contains multi-genre content, including drama, singing, and variety shows, reflecting media diversity. VOICES introduces controlled noise

Models	Training Data Domains
ECAPA-TDNN	Wideband, interview-style, augmented (speech, noise, music, babble, reverb, speed perturbation, compression)
TitaNET	Wideband, telephone, read speech, augmented (reverb, speed perturbation, time/frequency masking).
ECAPA2	Wideband, interview-style, augmented (speech, noise, music, babble, reverb, speed perturbation, time/frequency masking).
ReDimNet	Wideband, interview-style (VoxCeleb2), augmented (speech, noise, music, babble, reverb, speed perturbation))

Table 1: Domains used for training each speaker verification model.

conditions through additive distractors and far-field microphone placement. DIPCo captures conversational speech with mild overlap, simulating meeting environments. JuKeBOX consists of speech segments mixed with music, representing entertainment scenarios. These datasets provide complementary coverage of domain variations, enabling a comprehensive assessment of embedding robustness.

In addition to domain shifts, we analyze the impact of audio processing factors on model performance. We evaluate the effect of codecs by encoding audio samples with lossy and compressed compression formats. We also assess the impact of sampling rates by downsampling audio from 16kHz to 8kHz, followed by upsampling back to 16kHz, simulating frequency loss. Performance is measured using EER and minDCF in all experiments.

	CN-Celeb	VOICES	DiPCo	JuKeBOX
Ecapa-TDNN	21.66/0.87	6.13/0.35	9.15/0.5	34.62/0.80
TitaNet	18.07/0.83	10.43/0.68	3.38/ 0.21	28.30 /0.74
ECAPA2	16.93/0.68	4.22/0.22	3.63/0.26	29.27/ 0.71
ReDimNet	14.47/0.61	2.68/0.15	3.24 /0.29	30.93/0.73

Table 2: Speaker verification performance across four evaluation datasets from diverse domains: CN-Celeb, VOICES, DiPCo, and JuKeBOX. Results are presented as EER (%) / minDCF. Bold values indicate the best performance for each dataset.

5. Results and Discussions

5.1. Domain Sensitivity Across Datasets

Our evaluation shows that all tested models (ECAPA-TDNN, TitaNet, ECAPA2, and ReDimNet) experience performance degradation when evaluated on datasets representing domain shifts. As shown in Table 2, models trained on clean, wideband data achieve lower EER on CN-Celeb compared to DiPCo, VOICES, and JuKeBOX. This degradation is more pronounced in datasets characterized by far-field conditions, background noise, or music interference, indicating that current speaker embeddings overfit to training data, particularly clean, wideband speech.

Performance on VOICES highlights the sensitivity of embeddings to noise and reverberation, as the dataset includes noise augmentation and multi-microphone setups to emulate real-world acoustic variability. Models with poor results on VOICES are likely to struggle in practical applications involving distant-talking scenarios and non-ideal environments. A similar pattern is observed in DiPCo, which introduces conversational overlap and room effects, and JuKeBOX, which represents an extreme domain shift with speech mixed into music.

Among the models, ReDimNet achieves the lowest EER across most domain-shifted datasets, aligning with its focus on

improving robustness to domain variability. ECAPA2 outperforms ECAPA-TDNN in noisy and far-field conditions, such as VOICES and DIPCo, suggesting greater stability to acoustic variations. ECAPA-TDNN exhibits substantial degradation under these domain shifts, while TitaNet consistently yields the highest EER, indicating that its training regime and architecture are less effective in mismatched conditions. These results reinforce that architectural refinements alone are insufficient to ensure generalization, and embeddings remain vulnerable to realistic domain shifts.

	8khz	16khz
Ecapa-TDNN	5.94/0.48	3.79/0.27
TitaNet	4.71/0.39	2.33/0.16
ECAPA2	3.12/0.23	1.95/0.09
ReDimNet	2.69/0.32	1.83/0.08

Table 3: Speaker verification performance under different sampling rates: 8 kHz and 16 kHz. Results are reported as EER (%) / minDCF. Bold values represent the best performance in each condition.

5.2. Sampling Rate Variation

Performance degradation is observed across all models when reducing the sampling rate from 16kHz to 8kHz, as shown in Table 3. ReDimNet and ECAPA2 exhibit greater robustness, maintaining lower EER under both conditions. Although ECAPA-TDNN shows high absolute EER at 8kHz, its relative degradation is lower (57%) compared to TitaNet (102%), indicating that TitaNet is more sensitive to bandwidth reduction. These findings are consistent with prior work indicating that speaker embeddings depend on high-frequency information. The results emphasize that even state-of-the-art models are vulnerable to sampling rate limitations, which is critical in applications involving narrowband audio, such as telephone-based verification systems.

	WAV	MP3	FLAC	U-law
Ecapa-TDNN	3.79/0.27	5.28/0.41	3.79/0.27	4.10/0.28
TitaNet	2.33/0.16	3.30/0.28	2.33/0.16	2.41/0.17
ECAPA2	1.95/0.09	2.51/0.14	1.95/0.09	1.96/0.10
ReDimNet	1.83/0.08	2.29/0.13	1.83/0.08	1.86/0.09

Table 4: Comparison of speaker verification performance across different audio codecs. Results are reported as EER (%) / minDCF across four codecs: WAV (uncompressed), MP3 (lossy), FLAC (lossless), and u-law (companded). Lower values indicate better performance. Bold values highlight the best-performing model for each codec.

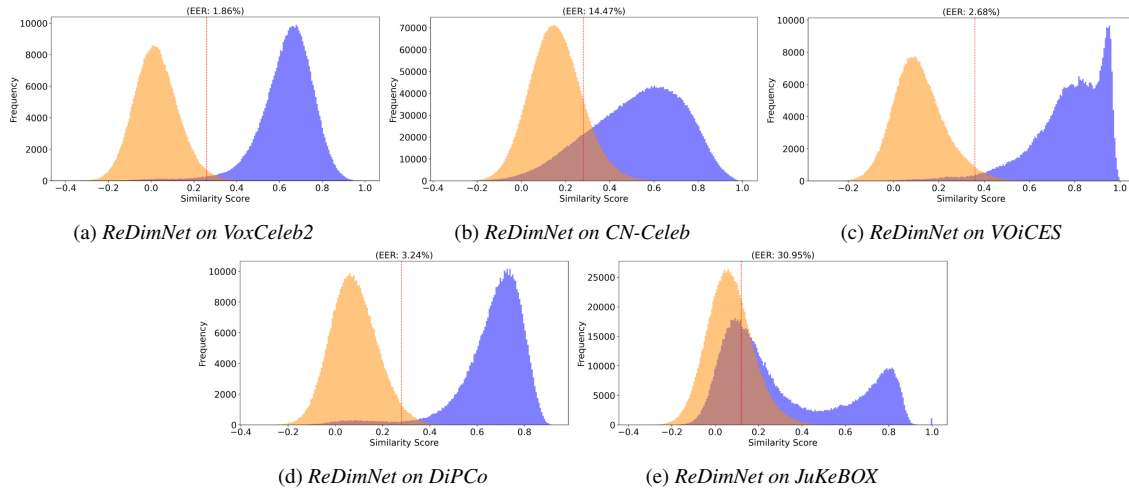


Figure 2: Distribution of cosine similarity scores for same-speaker (blue) and different-speaker (orange) trials across evaluation datasets using ReDimNet. The Equal Error Rate (EER) threshold is indicated by the dashed line in each plot. The spread and overlap of the distributions illustrate the verification difficulty across domains.

5.3. Codec Variation

Experiments with codecs reveal that all models degrade when audio is encoded using lossy or companded compression formats, as shown in Table 4. ReDimNet achieves the lowest EER across all evaluated codecs (WAV, MP3, FLAC, and u-law), suggesting greater robustness to compression artifacts. ECAPA2 is less affected than ECAPA-TDNN and TitaNet but is consistently outperformed by ReDimNet. ECAPA-TDNN exhibits the highest degradation, particularly in MP3 and u-law, indicating greater sensitivity to low-bitrate artifacts. TitaNet shows intermediate performance but also degrades under MP3 compression.

Although FLAC is a lossless codec, the source audio files were previously encoded using AAC before being transcoded to the tested formats. This pre-processing step likely explains why FLAC performance does not match expectations for truly lossless audio. These results reinforce that compression artifacts reduce the discriminative power of embeddings, especially in models more reliant on clean training data, and highlight that evaluation with seemingly lossless formats can still reflect encoding dependencies from upstream processing.

5.4. Score Distribution Analysis

To further analyze the behavior of the best-performing model, ReDimNet, we plot the cosine similarity score distributions for same-speaker and different-speaker pairs across evaluation datasets. Figure 2 shows that in challenging scenarios such as JuKeBOX and CN-Celeb, the distributions present greater overlap, aligning with the higher EERs observed. In contrast, DiPCo displays more separable distributions, corresponding to lower EERs.

However, VOiCES reveals an additional source of degradation: although the distributions may visually suggest separability, the application of multiple noise augmentation techniques and distractors introduces intra-speaker variability, causing samples from the same speaker to sound perceptibly different. This leads to increased false rejections, as the model frequently misclassifies same-speaker pairs as different speakers.

Visual inspection further indicates that music interference

in JuKeBOX often results in false acceptances, while reverberation and background noise in DiPCo primarily increase false rejections. This analysis complements the quantitative evaluation, reinforcing that domain conditions affect both the separability of embeddings and the consistency of same-speaker representations, particularly in noise-augmented datasets such as VOiCES.

6. Conclusion

This work investigated the sensitivity of deep speaker embeddings to domain shifts, sampling rate variations, and codec compression in speaker verification systems. We evaluated ECAPA-TDNN, ECAPA2, TitaNet, and ReDimNet across diverse evaluation conditions and observed performance degradation across all models when exposed to mismatched domains. ReDimNet demonstrated the lowest degradation percentage, indicating greater robustness, but it also exhibited limitations under severe shifts such as speech mixed with music.

Our results show that variations in sampling rates and codecs further degrade performance, reinforcing that speaker embeddings depend on high-frequency content and are vulnerable to compression artifacts. Even the most robust model, ReDimNet, experienced consistent but less severe degradation compared to others, suggesting that current embedding models remain susceptible to real-world audio variations despite architectural advancements.

These findings indicate that existing training paradigms relying on clean, wideband, and augmented data are insufficient for generalization to realistic conditions. Developing domain-invariant embedding models and adaptive training techniques that explicitly account for channel, noise, and sampling variability remains an open challenge. Addressing this sensitivity is critical to ensure reliable speaker verification performance in practical deployment scenarios involving heterogeneous audio environments.

7. Acknowledgements

This work has been fully/partially funded by the project Research and Development of Algorithms for Construction of Dig-

ital Human Technological Components supported by Advanced Knowledge Center in Immersive Technologies (AKCIT), with financial resources from the PPI IoT/Manufatura 4.0 / PPI HardwareBR of the MCTI grant number 057/2023, signed with EM-BRAPII.

8. References

- [1] D. Snyder, D. Garcia-Romero, G. Sell, D. Povey, and S. Khudanpur, "X-vectors: Robust dnn embeddings for speaker recognition," in *2018 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2018, pp. 5329–5333.
- [2] B. Desplanques, J. Thienpondt, and K. Demuynck, "Ecapa-tdnn: Emphasized channel attention, propagation and aggregation in tdnn based speaker verification," *arXiv preprint arXiv:2005.07143*, 2020.
- [3] N. R. Koluguri, T. Park, and B. Ginsburg, "Titanet: Neural model for speaker representation with 1d depth-wise separable convolutions and global context," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 8102–8106.
- [4] I. Yakovlev, R. Makarov, A. Balykin, P. Malov, A. Okhotnikov, and N. Torgashov, "Reshape dimensions network for speaker recognition," *arXiv preprint arXiv:2407.18223*, 2024.
- [5] A. Nagrani, J. S. Chung, and A. Zisserman, "Voxceleb: a large-scale speaker identification dataset," *arXiv preprint arXiv:1706.08612*, 2017.
- [6] W. Huang, B. Han, S. Wang, Z. Chen, and Y. Qian, "Robust cross-domain speaker verification with multi-level domain adapters," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 11 781–11 785.
- [7] Q. Wang, W. Rao, S. Sun, L. Xie, E. S. Chng, and H. Li, "Un-supervised domain adaptation via domain adversarial training for speaker recognition," in *2018 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2018, pp. 4889–4893.
- [8] T. Ko, V. Peddinti, D. Povey, M. L. Seltzer, and S. Khudanpur, "A study on data augmentation of reverberant speech for robust speech recognition," in *2017 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2017, pp. 5220–5224.
- [9] M. Van Segbroeck, A. Zaid, K. Kutsenko, C. Huerta, T. Nguyen, X. Luo, B. Hoffmeister, J. Trmal, M. Omologo, and R. Maas, "Dipco–dinner party corpus," *arXiv preprint arXiv:1909.13447*, 2019.
- [10] M. A. Pastor, A. Ortega, and D. Ribas, "Analysis of the domain mismatch problem in the speech emotion recognition task," in *Proc. IberSPEECH 2024*, 2024, pp. 181–185.
- [11] Y. Fan, J. Kang, L. Li, K. Li, H. Chen, S. Cheng, P. Zhang, Z. Zhou, Y. Cai, and D. Wang, "Cn-celeb: a challenging chinese speaker recognition dataset," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 7604–7608.
- [12] A. Chowdhury, A. Cozzo, and A. Ross, "Jukebox: A multilingual singer recognition dataset," *arXiv preprint arXiv:2008.03507*, 2020.
- [13] J. Thienpondt and K. Demuynck, "Ecapa2: A hybrid neural network architecture and training strategy for robust speaker embeddings," in *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2023, pp. 1–8.
- [14] L. Zhang, N. Jiang, Q. Wang, Y. Li, Q. Lu, and L. Xie, "Whisper-sv: Adapting whisper for low-data-resource speaker verification," *Speech Communication*, vol. 163, p. 103103, 2024.
- [15] D. Cai and M. Li, "Leveraging asr pretrained conformers for speaker verification through transfer learning and knowledge distillation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2024.
- [16] C. Richey, M. A. Barrios, Z. Armstrong, C. Bartels, H. Franco, M. Graciarena, A. Lawson, M. K. Nandwana, A. Stauffer, J. van Hout *et al.*, "Voices obscured in complex environmental settings (voices) corpus," *arXiv preprint arXiv:1804.05053*, 2018.